1.

Which of the following is one of the sorting algorithm?

A. Linear
B. Binary
C. selection
D. Greedy

Answer: C

2.

What is the best-case time complexity for Insertion Sort?

A. O(1)
B. O(n)
C. O(n log n)
D. O(n2)

Answer: B

3.

Which of the following sorting algorithms is generally preferred for very large datasets due to its stable O(n log n) performance?

A. Quick sort
B. Merge sort
C. Bubble sort
D. Insertion sort

Answer: B

4.

For an array that is already sorted, why does Insertion Sort achieve its best-case time complexity of O(n)?

A. The algorithm can detect that the array is sorted in the first pass and stops early.
B. It makes no swaps, so the time complexity is determined only by the number of elements.
C. Each element is compared only with the adjacent element to its left, requiring a single comparison per element.
D. It builds a heap data structure, which is faster to build with sorted data.

Answer: C

5.

For an array that is 'almost sorted' with only a few elements out of place, which algorithm would likely perform the fastest?

A. Selection sort
B. Merge sort
C. Heap sort
D. Insertion sort

Answer: D

6.

Which sorting algorithm uses recursion extensively?

A. Selection sort

B. Bubble sort

C. Insertion sort

D. Quick Sort

Answer: D